Question

The curve C has equation

$y = x^3 - 6x^2 + 9x + 2$

Find the coordinates of the stationary points of C and determine their nature.

Mark Scheme — CIE Style

Total: 7 marks

Mark	Requirement	Notes
M1	Differentiate y to obtain $dy/dx = 3x^2 - 12x + 9$	Method mark for using differentiation
A1	Correct derivative $dy/dx = 3x^2 - 12x + 9$	Accuracy mark; must be correct
M1	Set $dy/dx = 0$ and solve to find $x = 1$ and $x = 3$	Method mark for solving $3x^2 - 12x + 9 = 0$
A1	Correct x-coordinates $x = 1$ and $x = 3$	Accuracy mark
A1	Correct y-coordinates $y = 6$ when $x = 1$, and $y = 2$ when $x = 3$	Substitution into original equation
M1	Use d^2y/dx^2 or a sign test to determine nature	Method mark for applying a valid test
A1	Correct conclusion: maximum at (1, 6), minimum at (3, 2)	Accuracy mark; both points required

Notes:

- M marks require a valid method applied to this problem.
- A marks depend on the relevant M mark being earned or implied.
- If an earlier value is wrong but later working follows correctly, follow-through marks may be awarded where appropriate.